Toward a four-dimensional Yang–Mills construction

Notebook 34: ghost asymptotics and the physical determinant

Research working document

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Abstract

We determine the complete based-ghost contribution to the actual leading determinant's Maxwell coefficient. An exact returning-walk area calculation gives the unpinned lattice heat coefficient. Counting the root residues excluded by the pins controls the difference through a specified heat-time interval. Through the block diffusion time K^2, the actual contribution has a fixed finite part and an $O(K^{-2})$ error.

The flat pinning gap is of order K^{-4}, rather than K^{-2}. A pointwise Neumann Green-function estimate proves the matching lower bound. A fourth area moment and a killed-return heat bound then control the tail past K^4. Together they give the entire actual ghost coefficient $C_H \log K/(12\pi^2)+O(1)$, uniformly in the preceding classical depth and the permitted infinite structural systems.

This is twice the ghost logarithm of the massive reference used in Notebook 33. The extra logarithm comes from the interval K^2 to K^4 and must be compensated by the actual vector determinant. It supplies a quantitative test of the remaining coupled comparison, not the full Yang–Mills existence argument.

We also prove the full actual vector short-time bound, including the composite ordinary constraint. Its flat derivative has a uniform counting-ℓ^2 bound, and its affine magnetic jets are controlled through the fixed-ratio hierarchy. A localized trace identity uses the fixed coarse fiber dimension to avoid a factor equal to the growing fine block volume. Together with the ghost estimate, this closes the actual ultraviolet heat interval.

Finally an exact finite determinant identity separates the full fine and coarse gauge-orbit density factors from a residual precision on the physical quotient. That physical precision has a uniform flat gap of order K^{-2}. An explicit contour-gauge test shows why the same gap cannot simply be assigned to the unprojected contour completion. A paired scalar reference has an exact infrared cancellation; applying it to the actual finite identity still requires a joint infrared limit and uniform curvature-source estimates for the physical operator.

Inputs, conventions and scope

The actual construction, fixed-K local infinite limits and determinant-density identities are those of Notebooks 32 and 33. The input identities are:

1. On the normalized output probe $w_\mu(n) = tn_\nu H$, $\mu \neq \nu$, $\tau(H^2) = 1$, the first two actual stationary jets agree with current affine links $U_\mu(x) = \exp(itx_\nu H/K^2)$. This holds for every preceding homogeneous classical action S_M. Only these jets are used.

2. The based generator D_b has scalar domain zero at every K-center, and $G(t) = D_b(t)^T D_b(t)$. The Maxwell contribution of its normalized log determinant is $2[t^2] \operatorname{Tr}_{0,S} \log G(t)$, where the trace includes noncenter sites in one centered block and their adjoint color coordinates.

3. The actual total coefficient splits into this ghost term, the vector term $-[t^2] \operatorname{Tr}_{0,E} \log V_a(t)$, and a uniformly bounded coarse term. Here $V_a = s''_M + D_b D_b^T + aC_K^T C_K$ on the required jets; the

constraint multiplier curvature vanishes through the same two orders. The fixed a is the large positive completion from Notebook 32.

The charge factor is $C_H = \sum_q q^2$, where q runs over the eigenvalues of ad_H on the complexified adjoint space, with multiplicities. There is no doubling of the real adjoint trace. For $SU(N)$ with $\tau = \mathrm{Tr}_{\mathrm{fund}}/N$, $C_H = 2N^2$. We write $\alpha_g = C_H/(48\pi^2)$. All estimates below are in four dimensions.

The proofs have been checked internally against these inputs. This notebook does not assert a full endpoint-uniform vector comparison, positive total drift, a continuum measure, or reconstruction.

1 The actual ghost coefficient up to block diffusion time

Work in a joint infinite-lattice structural system at fixed K, with the based centers $K\mathbb{Z}^4$. Closed walks below return in $\mathbb{Z}^4$; no finite-torus winding or continuously varying periodic flux is used. The short-time closed-walk argument can be sharpened to compute the whole based-ghost contribution through heat time K^2 in current fine units. This is the complete short and intermediate interval for this term. It does not address the remaining vector or longer-time integral. Write

$$h_K^g(s) = -2[t^2]\,\mathrm{Tr}_{0,S}\, e^{-sG(t)}, \qquad \alpha_g = \frac{C_H}{48\pi^2}.$$

The stationary-jet identification of Notebook 33 makes this the same first two background jets of the actual based scalar operator for every preceding classical depth M. The field on a current link has curvature tH/K^2.

1.1 The unpinned diagonal coefficient

Let $X_1, \dots, X_4$ be independent continuous-time nearest-neighbor walks, each with jumps $+1$ and -1 at rate one. Their joint generator is the negative of the flat scalar lattice Laplacian. For the selected coordinate plane, set

$$\mathcal{A}(s) = \int_0^s X_\nu(u-)\, dX_\mu(u).$$

On a returning walk this is its oriented algebraic area, up to the immaterial orientation sign. Define

$$h_\infty^g(s) = C_H\, \mathbb{E}\left[\mathcal{A}(s)^2 \mathbf{1}_{\{X(s)=0\}}\right]. \tag{1}$$

The K^4 sites in one block cancel the curvature factor K^{-4}, so this is exactly the unpinned ghost contribution in the final-coordinate normalization.

For $s > 0$ define

$$I_0(2s) = \sum_{m\ge 0} \frac{s^{2m}}{(m!)^2}, \qquad I_1(2s) = \sum_{m\ge 0} \frac{s^{2m+1}}{m!(m+1)!}.$$

Proposition 1.1 (Exact lattice ghost heat coefficient)**.** *For every $s > 0$,*

$$h_\infty^g(s) = \frac{C_H}{3} s^2 e^{-8s} I_0(2s)^2 I_1(2s)^2. \tag{2}$$

Proof. A single coordinate returns precisely when its two Poisson counts of rate-one jumps agree. Its return probability is $p_s = e^{-2s} I_0(2s)$. Conditional on returning, let m be the count in either direction. The series above gives

$$\mathbb{E}[m \mid X_1(s) = 0] = s\, \frac{I_1(2s)}{I_0(2s)}.$$

Conditional on m, the m positive jump times and the m negative jump times are independent uniform points in $[0,s]$. It follows by integrating their signed point measures that, conditional on returning,

$$\mathbb{E}[X_1(u)X_1(v)] = v_s\{\min(u,v) - uv/s\},$$
$$\mathbb{E}[dX_1(u)dX_1(v)] = v_s\{\delta(u-v)\,du - du\,dv/s\}, \qquad v_s = 2\,\frac{I_1(2s)}{I_0(2s)}.$$

The second equality is an identity of second-moment measures, not a pointwise white-noise representation. For example, at fixed m its diagonal mass is $2m\,du/s$; the off-diagonal density is $-2m\,du\,dv/s^2$.

Conditioning the four coordinates to return preserves their mutual independence. There are almost surely no simultaneous jumps in different coordinates. Using the second-moment measure of X_μ and the bridge covariance of X_ν, therefore,

$$\mathbb{E}[\mathcal{A}(s)^2 \mid X(s)=0] = v_s^2\left\{\int_0^s (u-u^2/s)\,du - \frac{1}{s}\int_0^s\int_0^s \{\min(u,v)-uv/s\}\,du\,dv\right\}$$
$$= \frac{v_s^2 s^2}{12} = \frac{s^2}{3}\left(\frac{I_1(2s)}{I_0(2s)}\right)^2.$$

Multiplication by p_s^4 and C_H proves (2). All counts have finite moments, so these expectation interchanges are justified. □

Lemma 1.2 (Bounds with an integrable large-time error)**.** *There is a constant B, depending only on the color normalization, such that*

$$0 \le h_\infty^g(s) \le B\min(s^4,1), \qquad s>0, \tag{3}$$
$$|h_\infty^g(s) - \alpha_g| \le B/s, \quad |s(h_\infty^g)'(s)| \le B, \qquad s \ge 1. \tag{4}$$

Proof. The series give the small-s estimate. For $s \ge 1$ put $a_s = e^{-2s}I_0(2s)$, $b_s = e^{-2s}I_1(2s)$. Fourier coefficient extraction from $e^{s(e^{i\theta}+e^{-i\theta})}$ gives

$$a_s = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-2s(1-\cos\theta)}\,d\theta, \qquad b_s = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-2s(1-\cos\theta)}\cos\theta\,d\theta.$$

On this interval, with $\phi(\theta) = 2(1-\cos\theta)$,

$$\frac{4}{\pi^2}\theta^2 \le \phi(\theta) \le \theta^2, \qquad |\phi(\theta)-\theta^2| \le \theta^4/12.$$

Comparison with the full Gaussian integral gives

$$a_s = (4\pi s)^{-1/2} + O(s^{-3/2}).$$

Indeed the integrand error is bounded by $Cs\theta^4 e^{-cs\theta^2}$, whose integral is $O(s^{-3/2})$, and the Gaussian tail is smaller. The factor $\cos\theta - 1 = O(\theta^2)$ gives the same asymptotic formula for b_s. The integral formulas also give

$$|a_s| + |b_s| \le Cs^{-1/2}, \qquad |a_s'| + |b_s'| \le Cs^{-3/2}.$$

Substitute these estimates into $h_\infty^g = (C_H/3)s^2a_s^2b_s^2$. Its limiting value is $C_H/(48\pi^2)$; the relative product error is $O(s^{-1})$. Differentiating the product with the last bounds proves the stated derivative bound. □

1.2 Pinning changes this prefix by a vanishing amount

The centers form $K\mathbb{Z}^4$. A centered block contains exactly one representative of each residue class modulo K in each coordinate. For a closed walk shape γ of length $n \geq 1$ rooted at zero, let $r_K(\gamma)$ be the number of these root residues whose translated walk meets a center. Each vertex of the cyclic walk excludes one root residue at most, so

$$0 \leq r_K(\gamma) \leq \min(n, K^4). \tag{5}$$

This includes roots which themselves are centers. Algebraic area is invariant under translating a closed walk.

Proposition 1.3 (Uniform pinning comparison through time K^2)**.** *For every $s > 0$,*

$$0 \leq h^g_\infty(s) - h^g_K(s) \leq K^{-4}\{s(h^g_\infty)'(s) + 8sh^g_\infty(s)\}. \tag{6}$$

In particular, for $K \geq 1$,

$$\int_0^{K^2} |h^g_K(s) - h^g_\infty(s)| \, \frac{ds}{s} \leq B'K^{-2} \tag{7}$$

with B' independent of K, M.

Proof. The full closed-walk expansion and the actual center-avoiding expansion have nonnegative squared-area coefficients. Their difference is exactly

$$\frac{C_H}{K^4} e^{-8s} \sum_{n\geq 1} \frac{s^n}{n!} \sum_{\gamma:0\to 0,\ |\gamma|=n} r_K(\gamma)\mathcal{A}(\gamma)^2.$$

Inserting $r_K(\gamma) \leq n$ gives

$$\frac{1}{K^4} e^{-8s} s \frac{d}{ds}\{e^{8s} h^g_\infty(s)\},$$

which is (6). The quantity in braces on its right is nonnegative by this generating-series representation; no monotonicity of h^g_∞ alone is assumed. Integrating from zero to T yields

$$\int_0^T (h^g_\infty - h^g_K) \, \frac{ds}{s} \leq K^{-4} \left\{ h^g_\infty(T) + 8 \int_0^T h^g_\infty(s) \, ds \right\},$$

since $h^g_\infty(0) = 0$. The uniform bound (3), with $T = K^2$, proves (7). □

Theorem 1.4 (Actual ghost prefix with a fixed finite part)**.** *Define the finite constant*

$$\gamma_g = \int_0^1 h^g_\infty(s) \, \frac{ds}{s} + \int_1^\infty \{h^g_\infty(s) - \alpha_g\} \, \frac{ds}{s}.$$

Then the actual based-ghost contribution satisfies

$$\boxed{\int_0^{K^2} h^g_K(s) \, \frac{ds}{s} = \frac{C_H}{24\pi^2} \log K + \gamma_g + O(K^{-2}).} \tag{8}$$

The error constant is independent of the preceding classical depth M, K, and the permitted joint structural system.

Proof. Both integrals defining γ_g converge by Lemma 1.2. The tail of the second integral from K^2 to infinity is $O(K^{-2})$. Thus the same formula first holds with h^g_∞. Proposition 1.3 then proves it for the actual based ghost. Its independence of the effective action follows from the previously established affine stationary-jet identification. □

There is also the pointwise intermediate-scale bound

$$\left|h_K^g(K^2u)-\alpha_g\right| \le B''\{(K^2u)^{-1}+u/K^2\}, \qquad K^{-2}\le u\le 1. \tag{9}$$

Indeed (6) and the derivative bound give an error at most Cs/K^4 for $s\ge 1$, and the unpinned error is $O(s^{-1})$. This verifies the actual ghost part of the middle heat-time comparison with exponent one.

The remaining vector coefficient on that interval must supply $10\alpha_g$ in the continuum convention of Notebook 33. The estimate above does not bound $\int_{K^2}^\infty h_K^g(s)\,ds/s$ uniformly. At those longer times the pinning gap and coupled vector contribution must be retained. In particular, (8) is a theorem about a specified heat-time prefix, not a formula for the entire actual determinant or its sign.

2 The pinning scale and the entire based-ghost logarithm

The pinning scale is longer than the block diffusion time. We now control it sharply enough to determine the whole actual based-ghost Maxwell contribution. All statements retain the infinite-system and formal jet conventions of Section 1.

2.1 The correct flat pinned gap

Lemma 2.1 (Four-dimensional point-pinned Poincare bound)**.** *Let $G_K(0)$ be the flat scalar lattice Laplacian on functions vanishing at every site of $K\mathbb{Z}^4$. There is a constant $c>0$, independent of K, such that*

$$G_K(0)\ge cK^{-4}I. \tag{10}$$

The statement holds on compatible finite periodic volumes and on the infinite lattice.

Proof. It suffices to prove a block inequality using only edges internal to each centered cube. For $K=1$ the domain is zero; assume $K\ge 2$. The Neumann graph Laplacian of a path with K vertices has eigenvalues

$$\lambda_j = 4\sin^2\frac{\pi j}{2K}, \qquad 0\le j<K.$$

Its normalized cosine eigenvectors have squared values at most $2/K$. Products give an eigenbasis $\psi_{\boldsymbol{j}}$ in four dimensions, with

$$|\psi_{\boldsymbol{j}}(x)|^2\le 16K^{-4}, \qquad \lambda_{\boldsymbol{j}}=\sum_{\ell=1}^4\lambda_{j_\ell}\ge c|\boldsymbol{j}|^2/K^2.$$

Consequently the point diagonal of the zero-mean Neumann inverse is uniformly bounded:

$$\sum_{\boldsymbol{j}\ne 0}\frac{|\psi_{\boldsymbol{j}}(x)|^2}{\lambda_{\boldsymbol{j}}}\le CK^{-2}\sum_{\substack{\boldsymbol{j}\in\{0,\dots,K-1\}^4\\ \boldsymbol{j}\ne 0}}|\boldsymbol{j}|^{-2}\le C.$$

The last sum is $O(K^2)$, by shells in four dimensions.

Write a block field as $\phi=\bar\phi+u$, with u of mean zero. Its internal edge energy is denoted by $E_B(\phi)$. Spectral Cauchy–Schwarz gives $|u(x)|^2\le CE_B(\phi)$, and the first nonzero Neumann eigenvalue gives $\|u\|_2^2\le CK^2E_B(\phi)$. At the pinned center, $\bar\phi=-u(x_{\text{center}})$. It follows that

$$\|\phi\|_{2,B}^2=\|u\|_{2,B}^2+K^4|\bar\phi|^2\le CK^4E_B(\phi).$$

Sum over disjoint blocks. Internal edges are a subset of all edges, proving (10). The infinite statement follows first on finitely supported fields and then by closure. □

Together with the constant-off-centers trial function from Notebook 32, this gives the sharp order K^{-4} of the bottom of the flat based spectrum. The finite periodic upper test gives at most $8/(K^4-1)$; the same order follows in infinite volume by large cutoffs. It does not give a mass of order K^{-1}.

2.2 A killed-return bound and an area fourth moment

Let $p_s^K(x,y)$ be the heat kernel killed on visiting a center. At the flat background it is the kernel of $e^{-sG_K(0)}$. For $s \geq 1$, free-kernel domination and (10) imply

$$p_s^K(x,x) \leq Cs^{-2}e^{-cs/K^4}, \qquad x \notin K\mathbb{Z}^4. \tag{11}$$

Indeed split $e^{-sG_K/2}\delta_x$ into a time $s/4$ contraction followed by time $s/4$ smoothing:

$$p_s^K(x,x) = \|e^{-sG_K/2}\delta_x\|_2^2 \leq e^{-cs/(2K^4)}p_{s/2}^K(x,x) \leq e^{-cs/(2K^4)}p_{s/2}^{\text{free}}(0,0).$$

The last free return is $O(s^{-2})$ by the one-dimensional integral estimates already proved. Absorb the harmless factor two in c.

Lemma 2.2 (Fourth area moment of the returning walk)**.** *For $s \geq 1$, the free four-dimensional walk satisfies*

$$\mathbb{E}[\mathcal{A}(s)^4 \mid X(s) = 0] \leq Cs^4, \qquad \mathbb{E}[\mathcal{A}(s)^4 \mathbf{1}_{\{X(s)=0\}}] \leq Cs^2. \tag{12}$$

Proof. For a returning one-dimensional coordinate, its balanced jump count m satisfies the exact identity

$$\mathbb{E}[m^2 \mid X_1(s) = 0] = \frac{\sum_{m\geq 1} m^2 s^{2m}/(m!)^2}{I_0(2s)} = s^2.$$

Thus $\mathbb{E}m \leq s$ as well. Given the bridge path $Y = X_\nu$ and the count m for X_μ, its area is

$$\mathcal{A}(s) = \sum_{i=1}^m Y(U_i) - \sum_{i=1}^m Y(V_i),$$

where the U_i, V_i are independent uniforms in $[0,s]$, independent of Y. The deterministic mean terms cancel. The fourth moment formula for a sum of independent centered variables and Jensen's inequality give

$$\mathbb{E}[\mathcal{A}(s)^4 \mid m, Y] \leq C(m+m^2)\,\frac{1}{s}\int_0^s |Y(u)|^4\,du.$$

For clarity, centering $Y(U_i)$ changes its fourth moment by at most a fixed factor; the square of its second moment is at most its fourth moment.

Given the balanced count l of Y, the value $Y(u)$ is the difference of two independent Binomial$(l, u/s)$ variables. The same centered-sum formula bounds its fourth moment by $C(l+l^2)$, uniformly in u. Averaging counts therefore gives

$$\mathbb{E}[|Y(u)|^4 \mid Y(s) = 0] \leq C(s+s^2) \leq Cs^2.$$

The two coordinate bridges and their counts are independent. Combining the last bounds proves the conditional Cs^4 estimate. The four-dimensional return probability is $p_s^4 \leq Cs^{-2}$, proving the unconditional bound. □

Proposition 2.3 (Uniform ghost tail past the pinning scale)**.** *For $s \geq 1$,*

$$0 \leq h_K^g(s) \leq Ce^{-cs/K^4}. \tag{13}$$

In particular,

$$\int_{K^4}^\infty h_K^g(s)\,\frac{ds}{s} \leq C, \tag{14}$$

uniformly in K, M.

Proof. At each noncenter starting site x, the squared-area walk formula has the probabilistic form

$$h_K^g(s) = \frac{C_H}{K^4} \sum_{\substack{x \text{ noncenter} \\ x \text{ in block } 0}} \mathbb{E}_x \left[\mathcal{A}(s)^2 \mathbf{1}_{\{X(s)=x,\ X \text{ avoids centers}\}} \right].$$

Translate the area coordinate to x, which changes no area on a closed walk. Cauchy–Schwarz bounds each expectation by

$$\left(\mathbb{E}[\mathcal{A}(s)^4 \mathbf{1}_{\{X(s)=0\}}] \right)^{1/2} \left(p_s^K(x,x) \right)^{1/2} \le C e^{-cs/K^4},$$

using (11) and (12). The block has at most K^4 sites, canceling the prefactor. Integrate with $s = K^4 v$, $v \ge 1$, to obtain (14). □

2.3 The additional logarithm that the vector must cancel

Theorem 2.4 (Entire actual based-ghost coefficient)**.** *There is a constant C, independent of M, K and the permitted joint structural system, such that*

$$\left| \int_0^\infty h_K^g(s) \, \frac{ds}{s} - 4\alpha_g \log K \right| \le C. \tag{15}$$

More specifically,

$$\int_{K^2}^{K^4} h_K^g(s) \, \frac{ds}{s} = 2\alpha_g \log K + O(1), \tag{16}$$

and the tail past K^4 is uniformly bounded.

Proof. The proof of Proposition 1.3 gave, for every $T > 0$,

$$0 \le \int_0^T (h_\infty^g - h_K^g) \, \frac{ds}{s} \le K^{-4} \left\{ h_\infty^g(T) + 8 \int_0^T h_\infty^g(s) \, ds \right\}.$$

Set $T = K^4$. The right side is uniformly bounded, while the unpinned integral is $\alpha_g \log(K^4) + \gamma_g + O(K^{-4})$. This proves the full prefix law with an $O(1)$ error. Add Proposition 2.3 to get (15). Subtracting (8) gives (16). □

Recall explicitly that

$$\int_0^\infty h_K^g(s) \, \frac{ds}{s} = 2[t^2] \operatorname{Tr}_{0,S} \log G(t) = \beta(\mathcal{N}_0 \log \det G),$$

with the determinant-density and Maxwell normalization of Notebook 33. Thus the displayed factor two is the affine coefficient extraction, not an extra determinant. Its coefficient $4\alpha_g = C_H/(12\pi^2)$ is twice the $2\alpha_g = C_H/(24\pi^2)$ of the continuum ghost with mass K^{-1} in current fine units and the same ultraviolet scale. This difference is an actual pinning effect in the separated determinant; it is not a failure of the normalization.

In particular, a uniform estimate on the separated ghost tail starting at K^2 is false. Its positivity and (16) prove a logarithmic lower bound on that tail. The total reference coefficient is $22\alpha_g \log K$; hence a bounded comparison for the full actual determinant requires the vector contribution to be $18\alpha_g \log K + O(1)$, since the coarse determinant is bounded. Equivalently, the vector must compensate the additional ghost logarithm relative to its massive reference. No such vector estimate is established by this scalar calculation.

3 A uniform short-time estimate for the actual vector operator

Fix the primitive odd ratio $\ell \geq 3$, the actual selected analytic source mean, and $a > 0$. Let $K = \ell^k$, and let M be any preceding homogeneous depth. We work in an actual joint infinite structural system, with current-link angle counting coordinates. All affine substitutions below are polynomial jets at zero. No continuously varying periodic magnetic flux is used.

Choose $\mu \neq \nu$ and a Cartan element H with $\tau(H^2) = 1$, and write

$$U_\mu(t;x) = \exp(itx_\nu H/K^2), \qquad U_\lambda(t;x) = 1 \quad (\lambda \neq \mu).$$

The stationary-jet identification established in Notebook 33 gives, through order two, the actual vector family

$$\begin{aligned} B_M(t) &= s''_M(U(t)) + D_b(t)D_b(t)^T, \\ V_{M,K}(t) &= B_M(t) + aC_K(t)^T C_K(t). \end{aligned} \tag{17}$$

Here s''_M is the actual left-relative Hessian, D_b has scalar values zero at the K-centers, and C_K is the left-relative derivative of the composite ordinary map M_ℓ^k. In particular, s''_M is not replaced by the bare Wilson Hessian when $M > 1$.

The inputs used from the classical construction are uniform current-coordinate exponential bounds for $q_M = s''_M(1)$, the complete scalar-action support/Cauchy bounds through order four on a fixed complex product ball, and the exact local gauge covariance of the relative Hessian. The local coordinate and color fibers have fixed dimension. The actual primitive averaging formula supplies finite-range analytic derivative matrices. Its composite output row reads only the two final endpoint blocks. These inputs are uniform in preceding depth and compatible regulator volume.

Theorem 3.1 (Actual vector ultraviolet bound)**.** *There is a constant C, depending on the fixed primitive data, a, the classical analytic bounds and the color group, but not on M, K or the selected structural system, such that*

$$\int_0^1 \left| [t^2] \operatorname{Tr}_{0,E} e^{-sV_{M,K}(t)} \right| \frac{ds}{s} \leq C. \tag{18}$$

The trace is over fine bonds with initial site in block zero, including color. Together with the actual ghost ultraviolet bound, this proves the full short-time estimate on $0 < s < 1$, or $0 < u < K^{-2}$ after $s = K^2 u$.

3.1 Uniform flat bounds for the full composite constraint

Let D_r be the scalar forward difference on the lattice of spacing r, without division by r. Define the straight angle average

$$(A_r v)(y,\lambda) = r^{-d} \sum_{x \in B_r(y)} \sum_{h=0}^{r-1} v_\lambda(x + he_\lambda). \tag{19}$$

Let b_ℓ be the scalar mean of internal primitive root-contour sums. The actual arithmetic differential is

$$C_\ell = A_\ell - D_\ell b_\ell, \qquad C_\ell D_1 = D_\ell R_\ell, \tag{20}$$

where R_ℓ restricts scalar values to the new centers. Tiling centered blocks and concatenating strings gives $A_\ell^{(r)} A_r = A_{\ell r}$. Consequently

$$\begin{aligned} C_K &= A_K - D_K b_K, \\ b_{\ell r} &= b_\ell^{(r)} A_r + R_\ell b_r, \\ b_K &= \sum_{j=0}^{k-1} R_{\ell^{j+1} \to K}\, b_\ell^{(\ell^j)} A_{\ell^j}. \end{aligned} \tag{21}$$

Each scalar restriction in the last line is a counting L^2 contraction. At a final center it merely retains the same nested center; it does not sum all intermediate centers.

The absolute row sums of A_r equal r, and its absolute column sums equal r^{1-d}. For a column, the two neighboring longitudinal output intervals have complementary multiplicities summing to r, and the transverse output block is unique. Thus

$$\|A_r\|_{2\to 2} \le r^{1-d/2}. \tag{22}$$

These counting identities retain their multiplicities on compatible periodic graphs. Since b_ℓ is a fixed finite-range operator and $\|D_K\|_{2\to 2} \le 2\sqrt{d}$, in $d = 4$

$$\|C_K\|_{2\to 2} \le K^{-1} + \frac{2\sqrt{d}\,\|b_\ell\|_{2\to 2}}{1-\ell^{-1}}. \tag{23}$$

Every consecutive flat composite segment therefore has norm at most one fixed $C_0 \ge 1$, including empty segments. This is an $O(1)$ bound for the full ordinary map, not the generally false $O(K^{-1})$ assertion.

For localized information, a row of A_r has at most $2r^d$ entries, each at most r^{1-d}, so

$$\|A_r(c,\cdot)\|_2 \le \sqrt{2}\, r^{-1} \quad (d = 4).$$

Equation (21) gives uniformly bounded rows of b_K, and hence of C_K. The two-block dependency also shows that each input bond occurs in only a dimension-dependent number of different final output rows. This gives a second direct row-and-incidence proof of the full bound.

3.2 Actual affine jets of the nonlinear composite derivative

Fix first a final row with endpoint blocks near zero. At stage j, put $r = \ell^j$. The exact affine averaging identity gives the intermediate background angle, in that stage's lattice coordinates,

$$w_j(z) = t(r^2/K^2) z_\nu H.$$

The exact nested dependency geometry confines every needed coordinate to $|z| \le c_\ell K/r$. Therefore

$$\|w_j\|_{\text{needed}} \le c_\ell |t| r/K. \tag{24}$$

At each stage independently, extend the needed values to a globally bounded field with this same bound, by clipping the coordinate outside a larger dependency box. The extensions need not be successive blockings. Every derivative entry contributing to the selected product row agrees with the actual affine stage, which is the only use of these extensions.

Let F_j be the flat primitive derivative and $E_j(t)$ its change at the extended background. The fixed primitive analytic and incidence bounds give, on one disk $|t| < \rho_0$,

$$\|E_j(t)\|_{2\to 2} \le C_\ell |t| \ell^j/K. \tag{25}$$

Expand the derivative product by the subset of stages where E_j is used. A term with n marked stages has $n+1$ intervening flat segments, each bounded by C_0. Hence

$$\begin{aligned}\left\| \prod_j (F_j + E_j(t)) \right\| &\le C_0 \prod_j (1 + C_0 \|E_j(t)\|) \\ &\le C_0 \exp\left(C C_0 |t| \sum_{j<k} \ell^j/K \right) \\ &\le C_0 \exp(C C_0 |t|/(\ell - 1)).\end{aligned} \tag{26}$$

The actual selected row consequently has a uniform analytic L^2 bound. Cauchy estimates on a smaller fixed disk bound its Taylor coefficients through order two.

For a row at coarse position y, translating the affine field adds a constant Cartan potential. Remove it by the site gauge $\exp(ity_\nu x_\mu H/K)$. Exact gauge covariance expresses the row using its centered counterpart and input/output adjoint frames. On its two K-blocks the frame derivatives through order two are bounded by a fixed polynomial in $|y|$, independently of K. Writing $C_K(t) = \sum_j t^j C^{[j]}$, one may take

$$\|C^{[j]}(y,\cdot)\|_2 \le C_j(1+|y|)^{2j}, \qquad j = 0,1,2. \tag{27}$$

Their support remains in the same two endpoint blocks. This is a statement about polynomial jets; it does not assert a bounded analytic operator at a nonzero unbounded affine field.

Group current bonds by their initial-site K-block, and final bonds by their initial coarse site. The latter fiber has fixed dimension $r_c = d \dim \mathfrak{g}$. Equation (27) and finite incidence give finite-range block operator bounds with polynomial growth for $C^{[j]}$ and their transposes. In particular $C^{[0]}$ has a uniform weighted block Schur norm for every fixed block exponent.

3.3 The base vector before the constraint insertion

At zero,

$$B_0 = q_M + D_b D_b^T$$

is positive and has a uniform exponentially weighted fine-lattice Schur norm. A spectral gap is unnecessary on $0 \le s \le 1$: the Schur-algebra power series bounds its semigroup uniformly with exponential matrix decay.

The actual scalar-action bounds through order four give uniform exponentially rooted kernels for the first two background derivatives of B. The nearest-neighbor based-gradient term has the same property uniformly in the center spacing. For each fine diagonal bond rooted at x, use a Cartan gauge to subtract the constant potential $tx_\nu H/K^2$. The diagonal color heat trace is unchanged. Its source field is now $t(z_\nu - x_\nu)H/K^2$. Each linear magnetic vertex has a factor K^{-2}, and each quadratic vertex has K^{-4}; uniform exponential kernels absorb the remaining coordinate polynomials.

The one quadratic insertion and the ordered two linear insertions in the heat expansion therefore give

$$\left|[t^2]\,\mathrm{tr}_{\mathfrak{g}}(e^{-sB(t)})_{xx}\right| \le CK^{-4}(s+s^2)e^{Cs}, \qquad 0 < s \le 1. \tag{28}$$

The local gauge identities extend to these polynomial jets by their exponentially summable kernels. There are dK^4 bond roots per block, so

$$\left|[t^2]\,\mathrm{Tr}_{0,E}\, e^{-sB(t)}\right| \le C(s+s^2)e^{Cs}. \tag{29}$$

Its ds/s integral is uniformly finite.

3.4 Uniform block estimates without tracing the large fiber

The coefficients $B^{[j]}$, $j \le 2$, also have uniform exponentially decaying K-block operator kernels, with fixed polynomial growth in the block indices for $j > 0$. Fine positions in block X have magnitude at most $CK(1+|X|)$, while the magnetic source coefficients contain K^{-2j}. Fine Schur bounds control the operator norm of each block without its dimension as a prefactor. For distant blocks the fine distance is at least $K(d_{\mathrm{block}}(X,Y) - d)$; a smaller fixed block exponent is therefore summable uniformly in K. The finitely many nearby block offsets are controlled by the full fine operator norm.

Set $A(t) = aC_K(t)^T C_K(t)$. The preceding finite-range block bounds apply to $A^{[j]}$ and to $V^{[j]} = B^{[j]} + A^{[j]}$. For $E = B, V$, the block power series at zero and the one/two insertion formulas imply, for $j \le 2$,

$$\left\| ([t^j] e^{-sE(t)})_{XY} \right\| \le C(1 + |X| + |Y|)^p e^{-\kappa |X-Y|}, \qquad 0 \le s \le 1. \tag{30}$$

All constants are uniform in M, K. For positive j the usual factors s and $s + s^2$ can also be retained. Products of polynomial kernels are controlled by spending a smaller block exponent. No trace over a K^4-dimensional fiber has yet been taken.

3.5 The constraint correction has fixed coarse trace rank

Let P_0 project onto current bonds rooted in block zero. The exact Duhamel difference is

$$e^{-sV(t)} - e^{-sB(t)} = -a \int_0^s e^{-(s-u)V(t)} C_K(t)^T C_K(t) e^{-uB(t)} \, du. \tag{31}$$

At each coefficient through degree two, cyclicity gives

$$\mathrm{Tr}_{\mathrm{fine}}(P_0 E_V C^T C E_B) = \mathrm{Tr}_{\mathrm{coarse}}(C E_B P_0 E_V C^T). \tag{32}$$

The projector P_0 remains in this identity. The right side sums all coarse diagonal blocks; it is not identified with an arbitrary block-zero coarse trace density.

For each coarse root Y, its diagonal block on the right is a product through the fine block-zero Hilbert space. The outer fiber has dimension r_c, independent of K. Equations (27) and (30) bound the coefficients through degree two of the two maps by

$$C(1 + |Y|)^p e^{-\kappa' |Y|}.$$

The absolute trace of their product is at most r_c times the product of these norms. Summing Y is uniformly finite. Expanding the quadratic coefficient in (32), and then integrating the Duhamel time, proves

$$\left| [t^2] \, \mathrm{Tr}_{0,E}(e^{-sV(t)} - e^{-sB(t)}) \right| \le Cs, \qquad 0 < s \le 1. \tag{33}$$

One can first use finite boxes and compact sources. The displayed rooted sums are absolute and justify the limit. For fixed K, the finite P_0 also makes ordinary trace cyclicity legitimate; the estimate itself uses the small outer coarse fiber to avoid a K^4 factor. Complex coefficients use the holomorphic bilinear transpose, which agrees with the adjoint at the real flat point.

Combining (33) with (29) and integrating ds/s proves Theorem 3.1.

3.6 What this estimate does and does not settle

The argument uses the actual composite fixed-ℓ hierarchy. A direct-K contour average can have different raw row bounds and was not substituted. The full ordinary map needs only the proved $O(1)$ counting bound; sampled-center gauge directions forbid assuming $O(K^{-1})$ for it.

The final affine statement is in the infinite structural system. The uniform finite-regulator inputs and local limits do not turn it into a continuous magnetic-flux formula on an arbitrary torus. The semigroup estimates here use e^{Cs} on bounded time, not a uniform large-time gap. They prove neither the continuum coefficient on intermediate times nor the cancellation of the soft based-ghost contribution at large times. A combined signed-integral comparison remains an independent problem.

4 An exact finite physical quotient for the determinant

The identities in this section concern the actual finite homogeneous small-field germ. They remove the periodically pinned gauge variables before estimation. They do not identify the separate massless factors below with infinite-volume affine magnetic traces.

4.1 The actual spaces and the residual Hessian

Fix an old depth M, a composite endpoint $K = \ell^k$, and a compatible finite current torus. All matrices initially use real, trace-orthonormal color coordinates, current bond angles, and counting measure. Let E and E_c be the fine/current and final coarse bond spaces. Fix one final coarse site o as a common global gauge base. Let S_f, S_c denote the fine and coarse scalar site spaces whose value at o is zero. The noncenter scalar space S_b vanishes at every final coarse center. There is a coordinate splitting

$$S_f = S_b \oplus S_c, \qquad r : S_f \longrightarrow S_c, \tag{34}$$

where r is restriction to coarse centers; the coordinate extension of S_c is zero at noncenters.

At the actual fine saddle U and target W, write D_f, D_c for the left-relative gauge generators with these bases, and D_b for the restriction of D_f to S_b. Each generator with a global base is injective, by propagating a zero scalar value along lattice paths. The actual constraint differential $C : E \to E_c$ is onto, by its invertible pivot block, and exact gauge covariance gives

$$CD_f = D_c r, \qquad CD_b = 0. \tag{35}$$

Use the full relative-coordinate Lagrangian and coarse Hessians

$$L = s'' - \lambda \cdot f'', \qquad q = s''_{\mathrm{eff}}, \qquad \mathcal{H} = L - C^T q C. \tag{36}$$

The actual minimum-value comparison and its envelope identities give

$$\mathcal{H} \geq 0, \qquad \mathcal{H} D_f = 0, \qquad CH = I, \qquad \mathcal{H} H = 0, \tag{37}$$

where H is the derivative of the actual minimum section. The constraint-current term $\lambda \cdot f''$ is retained throughout. No on-shell hypothesis is required.

Let T be the actual contour-logarithm differential. Its normalization is $TD_b = -I$. Positivity of the actual retained precision implies that $\mathcal{H}$ on $\ker C$ has nullspace exactly $\operatorname{ran} D_b$ and is positive on its quotient. Indeed $\mathcal{H} = L$ on $\ker C$, and the contour component can be removed uniquely along D_b.

Define

$$G_f = D_f^T D_f, \quad G_c = D_c^T D_c, \quad P_f = I - D_f G_f^{-1} D_f^T, \quad P_c = I - D_c G_c^{-1} D_c^T. \tag{38}$$

Choose orthonormal column matrices U_f, U_c spanning the ranges of P_f, P_c, and put, for a fixed $a > 0$,

$$A = U_c^T C U_f, \qquad \mathcal{H}_\perp = U_f^T \mathcal{H} U_f, \qquad Q_\perp = \mathcal{H}_\perp + a A^T A. \tag{39}$$

These determinants do not depend on the orthonormal bases. The map A is onto: $P_f H U_c$ is a right inverse after passing to physical coordinates, since $P_c C D_f = 0$. Its range is annihilated by $\mathcal{H}$. It follows that $Q_\perp > 0$ and

$$A Q_\perp^{-1} A^T = a^{-1} I. \tag{40}$$

For example, if $H_\perp = U_f^T P_f H U_c$, then $A H_\perp = I$, $\mathcal{H}_\perp H_\perp = 0$, and $Q_\perp H_\perp = a A^T$, proving the last identity.

4.2 The exact determinant and orbit-density factors

Write $\mathcal{N}_0 F = F(W) - F(1)$, with the determinant branches continued from the positive flat convention. The actual leading correction obeys

$$\boxed{D_{M,K} = \mathcal{N}_0 \left\{ \frac{1}{2} \log\det G_f - \frac{1}{2} \log\det G_c - \frac{1}{2} \log\det Q_\perp \right\}.} \tag{41}$$

Both scalar determinants are essential. Empty determinants, when there is no nonbase coarse scalar coordinate, are one.

Here is a direct verification including the measure factors. Let K_0 be the actual retained insertion and C_p its pivot block. The leading amplitude is

$$I_{\mathrm{act}} = |\det C_p|^{-1} \det A_T^{-1/2}, \qquad A_T = K_0^T(\mathcal{H} + T^T T)K_0. \tag{42}$$

Its ambient constrained Gaussian is

$$\int_E dx\, \delta(Cx) e^{-\frac{1}{2}x^T\mathcal{H}x - \frac{1}{2}|Tx|^2} = (2\pi)^{(\dim E - \dim E_c)/2} I_{\mathrm{act}}. \tag{43}$$

Set $x = U_f u + D_f(\lambda_b, \lambda_c)$. The fine change of variables has Jacobian $\sqrt{\det G_f}$. The target basis $[U_c, D_c]$ has Jacobian $\sqrt{\det G_c}$. With $B = G_c^{-1} D_c^T C U_f$, one has exactly

$$Cx = U_c A u + D_c(Bu + \lambda_c), \qquad \delta(Cx) = (\det G_c)^{-1/2} \delta(Au)\delta(Bu + \lambda_c). \tag{44}$$

After integrating $\lambda_c = -Bu$, the residual action is $u^T \mathcal{H}_\perp u$. Since $TD_b = -I$, integration in λ_b is a translated standard Gaussian and contributes only $(2\pi)^{\dim S_b/2}$. In particular it produces no background-dependent pinned-site determinant.

The remaining physical constrained Gaussian, completed by aA^TA, has determinant factor $a^{\dim \operatorname{ran} P_c/2} \det Q_\perp^{-1/2}$ by (40). Since $\dim S_f = \dim S_b + \dim S_c$, its Gaussian powers agree exactly with (43). Thus

$$I_{\mathrm{act}} = \left(\frac{\det G_f}{\det G_c} \right)^{1/2} a^{\dim \operatorname{ran} P_c/2} \det Q_\perp^{-1/2}. \tag{45}$$

The a-only power disappears on applying $\mathcal{N}_0$, proving (41). The target delta uses a left-relative Lie coordinate. Its Haar density at the origin is constant; there is no additional $j(\log W)$ factor.

A useful basis-free ambient form is

$$Q_{\mathrm{res}} = \mathcal{H} + D_f D_f^T + aC^T P_c C, \qquad \det Q_{\mathrm{res}} = \det G_f \det Q_\perp, \tag{46}$$

because the full physical and vertical spaces reduce this operator. On the orthonormal vertical columns $D_f G_f^{-1/2}$ its matrix is G_f. Therefore

$$D_{M,K} = \mathcal{N}_0 \left\{ \log\det G_f - \frac{1}{2} \log\det Q_{\mathrm{res}} - \frac{1}{2} \log\det G_c \right\}. \tag{47}$$

The physical constraint $P_c C$ is unchanged by $C \mapsto C + D_c Z$. This observation applies after projection; it does not permit dropping the endpoint-gauge part of C in the original based ambient precision.

4.3 A direct canonical Schur check

For a second proof, start with the already established canonical completion

$$Q_b = \mathcal{H} + D_b D_b^T + C^T C, \qquad D = \mathcal{N}_0 \left(\log\det G_b - \frac{1}{2} \log\det Q_b \right). \tag{48}$$

Let $J : S_b \to S_f$ be the coordinate insertion, and in orthonormal full-gauge coordinates set

$$B_g = G_f^{1/2} J, \qquad C_{gg} = G_c^{1/2} r G_f^{-1/2}. \tag{49}$$

One has $C_{gg} B_g = 0$. Splitting the vertical space into $\operatorname{ran} B_g$ and its orthogonal complement, the first block contributes $\det G_b$. In the other vertical block C_{gg} is square and invertible. Its Schur elimination cancels the whole coarse-gauge component of CU_f, leaving $\mathcal{H}_\perp + A^T A$. Hence

$$\det Q_b = \det G_b \det(C_{gg} C_{gg}^T) \det Q_\perp\big|_{a=1}. \tag{50}$$

The scalar Schur identity for (34) gives $\det(r G_f^{-1} r^T) = \det G_b / \det G_f$. Consequently

$$\det Q_b = (\det G_b)^2 \frac{\det G_c}{\det G_f} \det Q_\perp\big|_{a=1}, \tag{51}$$

which independently checks every determinant power in (41). Changing a multiplies the physical determinant by $a^{\dim \operatorname{ran} P_c}$ and changes no normalized expression.

4.4 The finite identity and the infrared passage are distinct

All preceding identities hold on the actual finite real germ and extend bilinearly and holomorphically on a sufficiently small complex germ at that fixed regulator. The ambient form avoids choosing analytic orthonormal bases. No volume-independent complex neighborhood is inferred for the globally based scalar inverses.

The actual correction already has a local-density and affine polynomial-jet interpretation. The separate factors in (47) contain massless globally based gauges. The identity alone does not equate their separate infinite-volume magnetic traces to the actual Maxwell moment. A joint infrared-regulated or local-limit identity must retain all three factors. In particular, one may not replace G_f, G_c separately by massive translation-invariant ghosts without estimating the change.

4.5 A uniform flat physical gap

This subsection gives estimates for the actual flat operators, uniformly in M, $K = \ell^k$, and compatible volume in $d = 4$. The primitive ratio ℓ is fixed. The straight-string angle average is

$$(A_K^{\mathrm{st}} x)_\mu(y) = K^{-d} \sum_{z \in B_K(y)} \sum_{j=0}^{K-1} x_\mu(z + j e_\mu). \tag{52}$$

Its absolute row and column sums are K and K^{1-d}, respectively, including periodic multiplicities. Therefore $\|A_K^{\mathrm{st}}\|_{2\to 2} \le K^{1-d/2}$. The actual cumulative ordinary/corrected endpoint-gauge identity gives $C_K = A_K^{\mathrm{st}} + D_c Z_K$ at flatness. Thus

$$P_c C_K = P_c A_K^{\mathrm{st}}, \qquad \|P_c C_K\|_{2\to 2} \le K^{-1} \quad (d = 4). \tag{53}$$

No bound on Z_K is needed here. The same estimate must not be assigned to C_K before projection.

For the physical gap, take a flat physical field $x \in \operatorname{ran} P_f$ and put $\eta = K^{-1}$, $v = Kx$ in final physical units. Then

$$\|v\|_{2,\eta}^2 = K^{-2}\|x\|_2^2, \qquad P_c C_K x = P_c Q_K^{\mathrm{st}} v. \tag{54}$$

The four-dimensional action is unchanged by this rescaling. The previously proved curvature contraction gives, for the actual old-M flat quadratic energy,

$$\mathcal{E}_M(v) \geq \|d_\eta v\|_{2,\eta}^2. \tag{55}$$

The actual coarse Hessian q_{MK} is nonnegative, kills coarse gradients, and satisfies $\|q_{MK}\| \leq R_q$ with a common constant. The physical residual energy is therefore

$$\mathcal{E}_{\mathrm{res}}(v) = \mathcal{E}_M(v) - \langle P_c Q_K^{\mathrm{st}} v, q_{MK} P_c Q_K^{\mathrm{st}} v \rangle \geq 0. \tag{56}$$

Here is the needed uniform physical Poincare estimate. In each real coarse Bloch fiber p, use fine aliases $p + 2\pi r$, gradients $u_r = K(e^{i(p+2\pi r)/K} - 1)$ componentwise, and $P_r = |u_r|^2$. At flatness globally based and full periodic gradients have the same range, so $v_r \perp u_r$ is the correct physical condition. Write $d_c = e^{ip} - 1$, and let Ω_r be the diagonal normalized straight-average multiplier. The established elementary alias identities are

$$P_r \geq 4 \ (r \neq 0), \quad \sum_r |\omega_{r,\mu}|^2 \leq 1, \quad \min_\mu |\omega_{0,\mu}| \geq c_\omega = (2/\pi)^{d+1}, \quad \Omega_0 u_0 = a_0 d_c. \tag{57}$$

Set $y = P_{d_c}^{\perp} \sum_r \Omega_r v_r$ and $y_0 = P_{d_c}^{\perp} \Omega_0 v_0$. On the zero coarse mode the projection is I, including its harmonic one-forms. For a nonzero mode,

$$v_0 = P_{u_0}^{\perp} \Omega_0^{-1} y_0, \qquad |v_0| \leq c_\omega^{-1} |y_0|. \tag{58}$$

Indeed $\Omega_0 v_0 - y_0$ is a multiple of d_c, whose inverse image under Ω_0 is a multiple of u_0; orthogonal projection removes it. The zero-mode identity is simply $v_0 = y_0$. Parseval bounds the high-alias contribution to y by $\|v_{\mathrm{high}}\|$, and its energy is at least $4\|v_{\mathrm{high}}\|^2$. Consequently

$$\|v\|^2 \leq A_0 |y|^2 + B_0 \sum_r P_r |v_r|^2, \qquad A_0 = 2c_\omega^{-2}, \quad B_0 = \frac{1 + 2c_\omega^{-2}}{4}. \tag{59}$$

Summing fibers, using (55), and then (56), gives

$$\|v\|_{2,\eta}^2 \leq B_0 \mathcal{E}_{\mathrm{res}}(v) + (A_0 + B_0 R_q) \|P_c Q_K^{\mathrm{st}} v\|_2^2. \tag{60}$$

Thus the actual physical precision in (39) satisfies

$$\boxed{Q_\perp(1) \geq c_a K^{-2} I, \qquad c_a = \min\left\{ B_0^{-1}, \frac{a}{A_0 + B_0 R_q} \right\} > 0.} \tag{61}$$

This is uniform in refinement, endpoint, and finite volume. It supplies the expected physical gap after the gauge quotient. It does not give such a gap for the vertical block G_f, or bounds on derivatives of the nonflat projections P_f, P_c.

4.6 A limitation of the unprojected contour completion

For clarity, simply returning to the actual contour gauge does not give the gap (61) for its ambient matrix. First, the full flat ordinary average has a uniform $O(1)$ counting-ℓ^2 bound, though not the $O(K^{-1})$ bound of (53). Differentiate the exact cumulative gauge recursion. If b_ℓ is the one-step scalar endpoint contour mean, then, with nested center restrictions,

$$Z_K = \sum_{j=0}^{k-1} b_\ell^{(j)} A_{\ell j}^{\mathrm{st}}. \tag{62}$$

A row of b_ℓ has absolute norm at most $d\ell$. A row of A_r^{st} has squared norm at most $r\, r^{1-d} = r^{2-d}$, by its row sum and largest coefficient. Hence in $d = 4$ each Z_K row has norm at most $d\ell/(1-\ell^{-1})$ and each C_K row has norm at most $c_{\mathrm{row}} = 1 + 2d\ell/(1-\ell^{-1})$. Actual two-final-block locality makes each fine bond occur in at most $8d$ final rows. Rowwise Cauchy–Schwarz therefore proves

$$\|C_K\|_{2\to 2} \le c_C := \sqrt{8d}\, c_{\mathrm{row}}. \tag{63}$$

For $K \ge 3$, take one real color and a unit field e on the internal bond $[0, e_1]$ leaving one block center. Put $h = (K-1)/2$ and $\phi = T_K e$, with zero values at all block centers and outside this block. In the actual coordinate-order contour average, for $1 \le x_j \le h$ and $j \ge 2$,

$$\phi(1, x_2, \dots, x_d) = 1/d, \qquad \phi(0, x_2, \dots, x_d) = 0. \tag{64}$$

The root bond is traversed precisely when direction 1 is first among the d nonzero coordinate directions. Set $x = e + D_b\phi$. Exact gauge identities give $T_K x = 0$, $C_K x = C_K e$, and $x^T L_{\mathrm{flat}} x = e^T L_{\mathrm{flat}} e$. The h^{d-1} bonds in (64) differ from the original bond and have $|x| = 1/d$, so $\|x\|^2 \ge h^{d-1}/d^2$. The common flat operator bound and (63) imply

$$\lambda_{\min}(L_{\mathrm{flat}} + T_K^T T_K + a C_K^T C_K) \le d^2 (R_q + a c_C^2) h^{-(d-1)}. \tag{65}$$

In four dimensions this is $O(K^{-3})$, refuting a uniform lower bound cK^{-2} for that ambient contour completion. It is an upper Rayleigh bound, not a matching asymptotic. The physical quotient removes exactly the coordinate-gauge modes used by this test.

4.7 An infrared-cancelled scalar reference

The following is a specified infinite-lattice reference calculation. It is not yet a limit theorem for the globally based finite determinants in (41). Put $\alpha_g = C_H/(48\pi^2)$. The unpinned scalar magnetic heat coefficient established above is

$$h(s) = \frac{C_H}{3} s^2 e^{-8s} I_0(2s)^2 I_1(2s)^2, \quad h(s) = O(s^4)\ (s \downarrow 0), \quad h(s) = \alpha_g + O(s^{-1})\ (s \to \infty). \tag{66}$$

In common final physical heat time u, the fine and coarse coefficients are $h(K^2 u)$ and $h(u)$. Their orbit-metric reference ratio is exactly

$$\frac{1}{2}\int_0^\infty \{h(K^2 u) - h(u)\} \frac{du}{u} = \alpha_g \log K. \tag{67}$$

The integral is absolutely convergent. On $\epsilon < u < T$ its unhalved value equals $\int_T^{K^2 T} h(v)\, dv/v - \int_\epsilon^{K^2\epsilon} h(v)\, dv/v$; the endpoint limits prove (67). A common physical mass δ multiplies the integrand by $e^{-\delta^2 u}$, and dominated convergence gives the same limit as $\delta \downarrow 0$, for each fixed K.

This exact scalar cancellation is useful comparison data. Applying it to the actual correction still requires a joint infrared passage for (41), followed by uniform curvature-source estimates for the projected residual operator. The flat physical gap alone supplies neither assertion. Conditional on those further bridges, the physical term $-\frac{1}{2}\log\det Q_\perp$ would have to contribute $21\alpha_g \log K$ to obtain the target total $22\alpha_g \log K$; that contribution is not proved here.

5 The remaining coupled estimate

Let

$$B_v(M,K) = -[t^2]\,\mathrm{Tr}_{0,E}\log V_a(t).$$

The exact determinant identity, the bounded coarse contribution and Theorem 2.4 give

$$\beta(D_{M,K}) = B_v(M,K) + 4\alpha_g \log K + O(1).$$

The specified massive continuum reference has total coefficient $22\alpha_g \log K + O(1)$. Therefore a uniform bounded comparison with that reference is equivalent to the remaining estimate

$$B_v(M,K) = 18\alpha_g \log K + O(1)$$

over the full required family of preceding depths and structural systems.

If the actual vector prefix through time K^2 has its continuum value $20\alpha_g \log K + O(1)$, its remaining tail must be $-2\alpha_g \log K + O(1)$. This last statement is conditional on that vector prefix estimate. The scalar calculation alone gives no sign for the actual combined determinant. In particular a proof based on separately bounded vector and ghost tails past K^2 cannot work: the ghost tail has a proved logarithmic lower bound.

The full actual short-time interval $0 < s < 1$ is now bounded by Theorem 3.1 and the ghost estimates above. The unresolved analytic work lies at intermediate and longer times; it must retain the cancellation just identified.

The finite physical-quotient identity (41) provides another route to that combined estimate. It eliminates periodic pinning algebraically, keeps the fine/coarse orbit-density ratio, and exposes the flat physical gap (61). The next work on this route is to justify the joint infrared passage with all factors retained and to control the physical precision's curvature derivatives. Its spectral gap alone does not establish the required locality or the determinant coefficient.

Once a positive actual block margin is established with the needed uniformity, the finite drift-transfer and finite shooting results of Notebook 32 apply. The global measure matching, mixed and large-field histories, sourced continuum limits, nontriviality, Euclidean covariance, positivity and reconstruction remain requirements of the full existence objective.

References

[1] *Toward a four-dimensional Yang–Mills construction*, Notebooks 32 and 33 in this workspace. The actual determinant normalization, individually gauge-invariant densities, affine stationary jets and proper-time reference are used with their stated scope.

[2] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited accounts pinned at commit `e392ef45013a581664bab76b90867170b30fe10e`.